FINAL PROJECT REPORT

Utilizing Convolutional Neural Networks and Computer Vision for Traffic Sign Recognition

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1. Problem statement

For new drivers who have just gotten their license, there may be some traffic signs that they have not learned or encountered during their test. Some traffic signs may also be hard to understand without context or may have different designs that are not written in the guide for driving tests. This may lead to confusion when driving and may end up with unintentional violations of the rules. Traffic sign recognition has also been widely used in the creation of autonomous vehicles. Having good traffic sign recognition technology is essential for the vehicle to read road signs properly and help with driver safety [1]. Therefore, the need for a machine learning algorithm to detect and classify the traffic signs can help greatly with automation as well as provide easy access to learning. The use of Convolutional Neural Networks as the machine learning algorithm is due to its ability to detect objects in an image and classify or label what they are when trained. This mimics the ability of humans to recognise unique objects and understand what they are called once learned [11].

For the scope of this topic, we will only be using Philippine traffic signs in our dataset and will only be limited to the traffic signs inside this dataset. The algorithm used will be Convolutional Neural Networks (CNN) and Reinforcement Learning for machine learning in training. The product will only be the trained model and will only include various regulatory signs such as speed limit values signs, stop signs; traffic lights photos; bike lane signs; pedestrian crossings; and no turning signs. There will be no application for input of images and will remain as a trained model that can be integrated into an existing application.

2. Dataset

The researchers used one dataset entitled Philippine Traffic Sign Dataset by the user Jerry from Roboflow to train and test the model. The dataset contains 5.9k images and was split by 80% for training, 10% for testing and 10% for validation. The images include speed limit values signs (20-120), stop signs, red light, and green light. The images are captured in different situations, roads and distances. The photos also vary in quality, resolution, and angles.

The dataset cleaning was done using python by the removal of corrupted, duplicate, and small and low quality images. The dataset was preprocessed by standardizing the format to JPG RGB and the output is the number of images within the class folders.

The dataset split was also done using python and split into test, train, and validation folders. The libraries used to split the dataset were pytorch and tensorflow utilizing the cleaned dataset folder and creating a new folder called dataset_split.

3. Training

For the training of the models, the 80% split of the dataset was utilized to ensure accuracy when testing and validating. Python was used to train the CNN model and the pretrained CNN model used is MobileNetV2. The python libraries tensorflow which contains the pretrained model and matplotlib were used in training the CNN model. Reinforcement learning was implemented as a way to train the model and helped it reach accurate levels and was implemented using tensorflow and numpy. NLP was used for labelling and the libraries tensorflow, keras, numpy, nltk, and transformers. The NLP was utilized to fix the labels into human readable text and fixed the grammar.

4. Modeling

As stated in a study Convolutional Neural Networks (CNN) are a major component when it comes to Traffic Sign Recognition (TSR) Systems, these also act as a visual perception module of a system. Convolutional Neural Networks or CNN is a feed-forward neural network; it is designed to process visual data by enabling each neuron in every layer to respond simply to their immediate surroundings of the preceding layer [7]. CNN is a commonly used

algorithm when it comes to object detection rather than image classification. In image classification, the whole image is considered and labeled as such regardless of the environment. In object detection, the algorithm searches for the specific object in the image and classifies that object, enclosing it in a box for emphasis [11]. This is essential to traffic sign recognition as it is usually in the sides of roads where there may be multiple objects obstructing or can be distracting to the eye. This is why CNN was the chosen algorithm for this specific project.

The Convolutional Neural Networks (CNN) has a different component, which are the building blocks when it comes to understanding the input images/symbols in the Traffic Sign Recognition (TSR) Systems. *Convolutional layer* as the first component, is responsible when it comes to identifying a low-level patterns/feature extraction. While the next layer would be the *activation layer*, the main function of this is to present non-linearity into a neuron's output. Based on the study it will determine whether a neuron should be activated or not, this can be achieved by calculating the weighted sum of the inputs and adding the bias. Third is the *pooling layer*, this will reduce the physical proportion of the feature maps while keeping their most necessary features from the window. Lastly is the *fully connected layer*. This is the one that will be responsible for the final predictions based on the high-level features that have been learned by the previous layers [8].

The specific CNN model used was MobileNetV2 which is a 53-layer deep lightweight CNN model and was developed by the integration of residual and linear bottleneck modules into MobileNetV1 [12]. This model has two convolution layers which are depthwise and pointwise convolution. In the depthwise layer, a single convolution is applied to each input channel to filter it. The pointwise layer with a kernel size of 1x1 computes a linear combination of input channels which can reach sparse representation. This architecture represents the process of expansion, convolution feature extraction, and compression [13].

Minimum Viable Product, or MVP, is a product with essential features to attract users' needs, it is developed and released to quickly gather feedback from the early adopters. This has become a strategic tool in agile development because it allows teams to validate their ideas, enhance production-market fit, and make decisions based on the real data of its user [9].

There are five stages in MVP: empathize, define, ideate, prototype, and test. Throughout these stages, the developers would be involved with users directly for them to understand users' experiences, challenges, or frustrations to be able to create solutions that will connect with the users [10].

To train the CNN algorithm, the Reinforcement Learning (RL) approach was utilized. In reinforcement learning, an agent has a state and can receive a reward from the environment. The goal of the agent is to determine an appropriate action after a reward is given. The agent can either maximise the given reward or exploit an action. Although, solving problems through exploitation can sometimes lead to reward sparseness. To fix sparse reward problems, exploration is used as a solution. Exploration is the act of searching for a reward as a solution to the problem. There are many different exploration solution types that exist, but in this project, a random exploration is utilized wherein the agent decides to do it randomly regardless of its progress [14]. In our system, the agent which is the CNN model, has a state where they are labelling the inputted image and the reward is if the labelling of this image is correct they get a point.

For the labelling of the image, an NLP algorithm is used. In an NLP system, a natural language is converted into a class type for classification tasks, a sequence of labels for sequence-labeling tasks, or a different sentence for generation tasks. For our system, it holds a sequence-labeling task wherein the CNN provides a label and the NLP translates this label into readable text. CNNs are usually used for tasks such as word embedding due to the fact that CNNs contain feature extraction which can provide the NLP algorithm with context of the images [15].

5. Evaluation

The model shows that it is well-calibrated, especially at high confidence levels (see Figure 1). Some miscalibration exists in the mid-confidence range. High-confidence predictions can be more trusted than moderate-confidence ones.

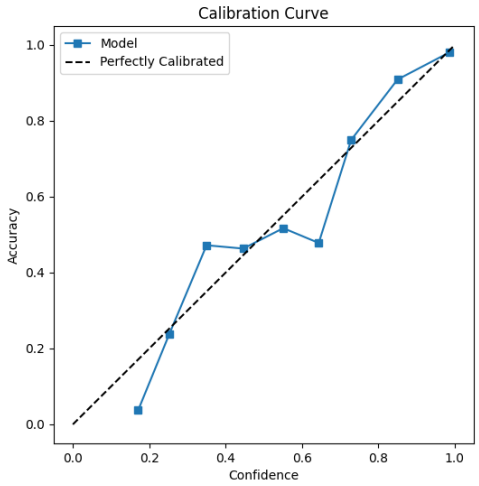


Figure 1: Calibration Curve

The ablation study compares three configurations: the full model, a version without dropout, one without data augmentation (see Figure 2). The full model achieves the highest accuracy. When dropout is removed, accuracy drops significantly. It suggests that dropout plays a crucial role and helps prevent overfitting. Lastly, removing data augmentation also reduces accuracy. This indicates that augmentation should expose the model to more diverse training examples.

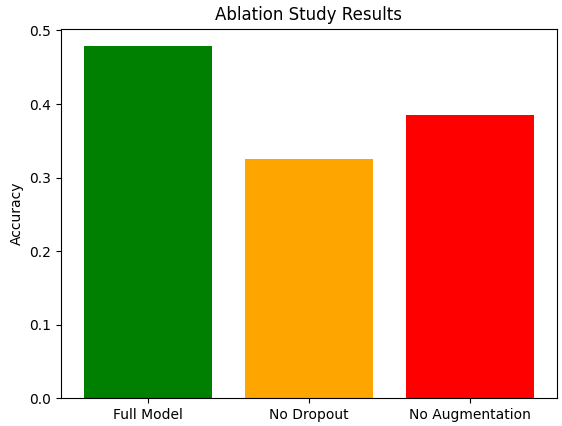


Figure 2: Ablation result

The confusion matrix shows how the model behaves across different classifications beyond overall accuracy (see Figure 3). It shows that a stop sign has the strongest performance, showing the model clearly distinguishes it. Some speed limits perform well but others do not. In terms of green light and red light, it performed with moderate accuracy .

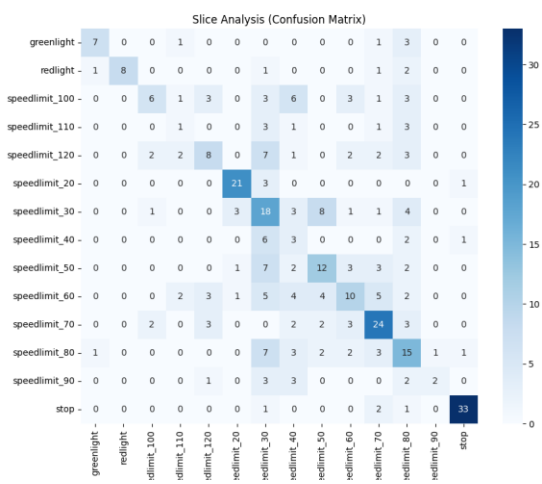


Figure 3: Confusion Matrix

The learning curve below shows how the model's performance evolves during training, using both accuracy and loss for the training. For accuracy, both training and validation increase overtime. The two curves stay relatively close, which suggests the model is not severely overfitting. There are small fluctuations in validation accuracy. In terms of loss plot, both training and validation loss decrease consistently. This shows the model is improving and converging without major instability.

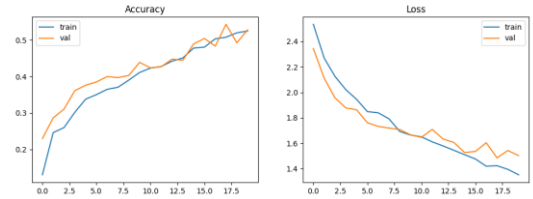


Figure 4: Training History Plot/Learning Curve

6. Ethics & Policy

The ethical considerations of this study focus on the responsible development and evaluation of artificial intelligence models. The proposed traffic sign classifier using Convolutional Neural Networks and reinforcement learning is evaluated within a controlled simulation environment and is intended for academic and experimental purposes. The dataset used in the study is limited to traffic lights, speed limit signs, and stop signs, which means the model represents a specific set of traffic sign categories and conditions included in the dataset. The results therefore reflect the performance of the model within the defined scope of the experiment and dataset. Any future implementation in real-world applications would require additional validation, expanded datasets with more diverse traffic signs and conditions, and further evaluation procedures to ensure reliable and safe performance.

6.1 Intended Use

The traffic sign recognition system is designed to demonstrate the application of CNN in image classification. CNN learns patterns like edges, shapes, and colors from traffic sign images, then uses those patterns to classify signs. It also supports academic learning and experimentation in computer vision in which it allows users to adjust model parameters, and observe how accuracy changes. This web app also serves as a prototype or a sample model for traffic sign recognition, it is not yet fully optimized. Lastly, it is not intended for real-world or safety-critical deployment. It may fail under different lighting, weather, or angles. It also lacks safety validation.

6.2 Limitations

The traffic sign recognition has limitations including limited dataset, dataset-dependent, poor generalization capability. First, the model is trained on a limited dataset, which only represents a few real-world traffic signs. As a result, it does not fully capture the variability present in actual driving environments. Additionally, the model's performance metrics are dataset-dependent, meaning its accuracy and reliability are only validated within the scope of the dataset used during training and testing, which restricts the validity of the results. Furthermore, the model may exhibit poor generalization capability, as it has not been exposed to variation of environmental conditions. Consequently, its predictive performance may significantly degrade when applied outside the training context.

6.3 Ethical Risks

The first risk is the misclassification, where incorrect predictions could lead to unsafe interpretations if the model is used beyond its intended academic purpose. There is also a risk of overgeneralization, wherein users mistakenly assume that the model is robust enough for real-world deployment. This misunderstanding can lead to inappropriate reliance on the system.

Lastly, safety concerns arise if the model is integrated into real-world traffic systems without validation. Minor inaccuracies could have serious consequences, emphasizing the importance of responsible use.

6.4 Risk Mitigations

To address these risks, several mitigation measures are implemented. The system is explicitly labeled as academic and experimental, ensuring that users clearly understand its intended purpose. An ethics statement documentation is provided to highlight all known limitations and constraints. Moreover, the project avoids making any claims regarding real-world applicability. Importantly, users are explicitly warned against deploying the system in safety-critical environments.

6.5 Responsible Use

Users of the Traffic Sign Classifier are expected to adhere to responsible usage guidelines. The system should be utilized strictly for educational, research, and experimental purposes. Users must refrain from deploying the model in real-world applications without conducting extensive validation, testing, and optimization.

6.6 License

This project is distributed under the MIT License, an open-source license that allows users to use the system freely. Specifically, users are allowed to use, modify, copy, and distribute the system.

6.7 Fairness

The model may perform unevenly across different types of traffic signs. For instance, it may recognize some signs more accurately than others due to dataset biases. It is crucial to acknowledge potential bias and avoid assuming universal accuracy.

7. Model Card

Field	Details
Model	**Model Name:** Traffic Sign Recognition **Algorithms used:** CNN, NLP, RL **Objective:** To develop a model that can detect and classify different traffic signs from images input.
Architecture	MobileNetV2
Dataset	The dataset that will be used for this project proposal is https://universe.roboflow.com/jerry-imbhm/philippine-traffic-sign-dataset-9kz1e-qvpnr The dataset only includes stop sign, red light, green light, and speed limits images (20-120)
Data Splits	**Train:** 80% **Test:** 10% **Validation:** 10%
Cleaning	● Removed corrupted, duplicates, and small images. ● Images are converted to jpg.
Intended Use	This system is not intended for deployment in real-world or safety-critical environments, such as autonomous driving systems, without extensive further development and validation.
Limitations	● The model represents only a narrow subset of real-world traffic conditions ● The system may not generalize well to different environments, lighting conditions, or unseen traffic signs ● Performance results are limited to the dataset used
License	MIT License

8. How to run

To run the project, the user needs to ensure that the environment that they are using has Python 3.8, Python 3.11, or higher installed as well as the other packages that are crucial for the program to run, such as TensorFlow, Pillow (PIL), and Tkinter libraries. Users will

need to prepare their environment by running their cleaning file (cleaning.py) that they downloaded, this will remove the corrupted or duplicated files in your data. The users need to ensure that the model files are inside the folder named "model" so that the application can locate the files.

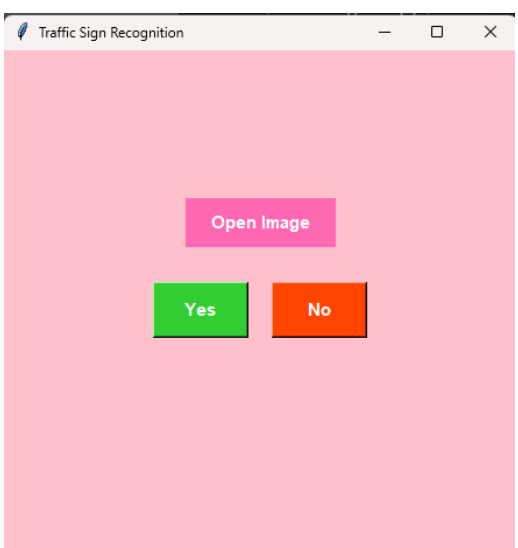
```
PS C:\Users\Admin\Downloads\traffic-sign-classifier-main> & C:/Users/Admin/AppData/Local/Programs/Python/Python311/python.exe c:/Users/Admin/Downloads/traffic-sign-classifier-main/traffic-sign-classifier-main/data/cleaning.py

Corrupted removed: 0
Duplicates removed: 0
Small images removed: 0
Converted to JPG: 0

=== CLASS COUNTS ===
greenlight : 74
redlight : 82
speedlimit_100 : 164
speedlimit_110 : 56
speedlimit_120 : 176
speedlimit_20 : 163
speedlimit_30 : 255
speedlimit_40 : 76
speedlimit_50 : 193
speedlimit_60 : 235
speedlimit_70 : 255
speedlimit_80 : 228
speedlimit_90 : 67
stop : 243
```

After cleaning the data, the path is right, as well as the files. The user can now run the interface of the traffic sign recognition by executing the entrypoint.py.

```
PS C:\Users\Admin\Downloads\traffic-sign-classifier-main\traffic-sign-classifier-main> & C:/Users/Admin/AppData/Local/Programs/Python/Python311/python.exe c:/Users/Admin/Downloads
/traffic-sign-classifier-main/traffic-sign-classifier-main/models/entrypoint.py
```



To evaluate the models/finetune.py the user will need to run the file. The user will see the model: sequential and after running the epoch, users will be able to see the test accuracy, and different figures such as Calibration Curve, Confusion Matrix, and Ablation result.

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 62, 62, 32)	896
max_pooling2d (MaxPooling2D)	(None, 31, 31, 32)	0
conv2d_1 (Conv2D)	(None, 29, 29, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 14, 14, 64)	0
conv2d_2 (Conv2D)	(None, 12, 12, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 6, 6, 128)	0
flatten (Flatten)	(None, 4608)	0
dense (Dense)	(None, 128)	589,952
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 14)	1,806

```
Total params: 685,006 (2.61 MB)
Trainable params: 685,006 (2.61 MB)
Non-trainable params: 0 (0.00 B)
Epoch 1/20
50/50 ━━━━━━━━━━━━━━━━━━━━ 4s 65ms/step - accuracy: 0.1436 - loss: 2.5291 - val_accuracy: 0.2806 - val_loss: 2.3361
Epoch 2/20
50/50 ━━━━━━━━━━━━━━━━━━━━ 3s 60ms/step - accuracy: 0.2302 - loss: 2.2789 - val_accuracy: 0.2925 - val_loss: 2.1330
Epoch 3/20
50/50 ━━━━━━━━━━━━━━━━━━━━ 3s 61ms/step - accuracy: 0.2663 - loss: 2.1377 - val_accuracy: 0.3433 - val_loss: 1.9991
Epoch 4/20
50/50 ━━━━━━━━━━━━━━━━━━━━ 3s 60ms/step - accuracy: 0.3080 - loss: 2.0208 - val_accuracy: 0.3612 - val_loss: 1.8861
Epoch 5/20
50/50 ━━━━━━━━━━━━━━━━━━━━ 3s 60ms/step - accuracy: 0.3163 - loss: 1.9788 - val_accuracy: 0.3493 - val_loss: 1.8873
```

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Ozoudeh/publication/387008612_Best_Practices_in_Project_Management_for_Technology-_Driven_Initiatives_A_Systematic_Review_of_Market_Expansion_and_Product_Development_Techniques/links/675c001ada24c8537c6a1761/Best-Practices-in-Project-Management-for-Technology-Driven-Initiatives-A-Systematic-Review-of-Market-Expansion-and-Product-Development-Techniques.pdf

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Team roles & milestones

Members	**Roles**
Bacani, Cyril	Data & Ethics Lead, CNN Components and Minimum Viable Product
Egaran, Chelsea	Modeling Lead,Dataset, Ethical Considerations
Magpantay, Sheena	Project Lead / Integration, Review of Related Literature
Manalo, Kisha	Evaluation & MLOps Lead, Problem Statement and Scope